BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Final | Semester: Fall 2025
Duration: 2 hours | Full Marks: 55

CSE 320/EEE361/ECE361: Data Communications

Answer the following questions.
Figures in the right margin indicate marks.

SET B

Name:	ID:	Section:

1. [CO2]

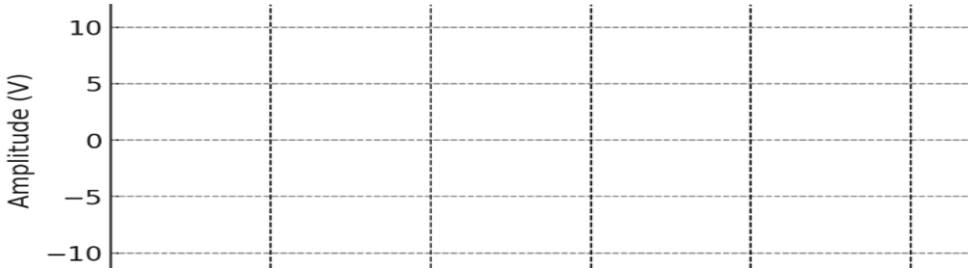
a) A 4-FSK communication system has the following parameters: [1+2+2]

- Carrier frequency f_c = 4 Hz
- Frequency deviation Δf = 2 Hz
- Amplitude: 5V

i. **Determine** the number of bits per signal.

ii. **Find** the frequencies used for the 4-FSK (M-FSK), if the frequencies are determined by the formula $f_k=f_c+(k-(M+1)/2)\Delta f$, where k=1,2,3,4 and M is the number of frequencies **[Hint: k=1 refers to 00, k=2 refers to 01 and so on]**

iii. **Draw** the transmitted FSK signal in the graph given below for the following data: **10000100**



b) **Explain** why digital data must be converted into analog signals for certain transmission media. What role does the carrier signal play in this process? [3]

c) **Describe** how Quadrature Phase Shift Keying (QPSK) increases data rate without increasing bandwidth. [2]

2. [CO3]	a)	**Six** analog signals are multiplexed using **Frequency-Division Multiplexing (FDM)** to transmit over a single communication link. The bandwidth requirements of the channels are as follows: • Channels 1–4: 160 kHz each • Channels 5–6: 240 kHz each A guard band of 15 kHz is inserted between every pair of adjacent channels. i. **Determine** the total bandwidth required for the multiplexed FDM signal if the initial frequency starts from 401 kHz. ii. **Draw** the spectrum of the FDM signal. iii. Suppose the guard band is reduced from 15 kHz to 10 kHz. **Briefly explain** the impact on the received signal and the data rate.	[1+1 +2]
	b)	Consider a system with 16 source channels, each generating data at a rate of 4000 bits per second. The channels are combined using **Statistical Time Division Multiplexing (TDM)** with character interleaving. Since not all sources are active at the same time, each frame carries data from 6 active sources. Each slot in the frame requires an additional 3-bit address field to identify the source. i. **Determine** the size of the output frame in bits. ii. In which scenario, synchronous TDM is efficient over statistical TDM. Briefly justify your answer.	[2+2]
	c)	A Time-Division Multiplexing (TDM) system consists of m input lines. The first two input lines operate at a data rate of 16 kbps each, the third input line operates at 27 kbps, and all remaining input lines operate at 32 kbps each. The interleaving unit is 2 bits. In addition, synchronization overheads of 2 bits per frame is used. The output data rate is 288 kbps. Under these constraints, i. **Draw** the figure to visualize the multiplexing process with proper labels. ii. **Calculate** the input slot duration. iii. **Find** the Frame rate. iv. **Determine** the Frame size. v. **Identify** the number of input lines that can be supported.	[2+1 +1+1 +2]

3. [CO4]	a)	A sender transmits the code word 1010110010. Using the notation **C(10,7)**, **find** the corresponding **divisor.** If Cyclic Redundancy Check (CRC) is used at the receiver side to detect errors, **determine** whether the received code word 1011110010 contains any error. (**Hint:** In the divisor, the Least Significant Bit (LSB) is 0 and all other bits are 1.)	[2+3]
	b)	A linear block coding scheme is used to encode Datawords into Codewords. The following table shows the encoding scheme: Dataword / Codeword 00 / 0000000 01 / 0111100 10 / 1000111 11 / ? Note: In a linear block coding scheme, XOR-ing two distinct valid codewords produces another distinct codeword. • From the table above, **find** the missing codeword. • **Determine** the maximum number of errors this scheme can detect and correct. • The receiver received the codewords 1011010 and 1100111. **Determine** if the original dataword can be retrieved from it. If so, what are the original datawords? **Show** detailed calculations.	[1+3 +3]
	c)	You are given the following four 16-bit Checksum calculations at a sender device, where some of the values are missing. Your job is to analyze the calculations and answer the following questions. All the words are 16-bits and represented in Hexadecimal values. You **must** also answer in **Hexadecimal values**. For simplicity, the fake checksum (word with all zeroes) has been omitted in the summation process. BAE — First Data FACE — Second Data **?** — Third Data BEEE — Fourth Data (+) C048 — Initial Sum 2 — Sum wrapping value (+) **?** — Wrap Sum **?** — Final Checksum	[2+2]

		i. What is the actual value for **Warp Sum** and **checksum**? ii. What is the actual value for **Third Data**?	
4. **[CO4]**	a)	In a smart city parking system, hundreds of parking sensors are installed across different parking slots. Each sensor sends availability updates to a central server using a shared communication channel. When a sensor has data to send, it first listens to the channel. If the channel is idle, the sensor transmits instantly. If the channel is busy, the sensor does not keep sensing the channel. Instead, it waits for a random time before sensing the channel again. i. Name which CSMA variant (1 persistent, non-persistent or p-persistent) is used here. ii. Discuss a major issue with the variant mentioned in (i) iii. CSMA protocol cannot detect collisions. Name the protocol that can detect a collision. Explain what happens after a collision is detected.	[1+2 +2]
	b)	In a shared communication link, there are two stations on either side. They are using the CSMA/CD protocol to detect any collision occurring during transmission. The minimum frame size to detect collision is 1200 bytes, the speed of propagation is 5 x 10^7 m/s, and the bandwidth of the channel is 30 Mbps. i. **Calculate** the minimum transmission time to detect a collision. ii. **Find out** the vulnerable time of the system. iii. **Identify** the distance between the two stations.	[2+2 +2]
	c)	A disaster management authority uses a **satellite-based communication system** to receive messages from **hundreds of remote emergency stations** deployed across coastal and mountainous regions. All stations share the same communication channel to send alert messages to the satellite. During normal conditions, the satellite introduces a **global timing signal** and divides time into **fixed-length slots**. Emergency stations are instructed to transmit messages **only at the beginning of a time slot**, following the synchronization signal. The frame duration is 48.6 μs and the propagation delay is 170 ms for any message transmitted. i. **Explain** with proper reasoning which technique is used in this case? ii. **Calculate** the vulnerable time for the communication system.	[2+1]

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